Saatwik Sairaam Vasamsetti

Full-Stack Web Developer · Software Engineer

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PROFILE

Full-stack engineer who ships production-shaped systems and measures them. Recent work spans Trade Research, where Python computes every figure and a verifier traces each one in a generated report back to its source data, and an offline-first emergency platform whose TypeScript/PostGIS API passes 636 assertions with a Kotlin Compose client and a YOLO11 pipeline beside it. Comfortable across the stack — API and schema design, transactional concurrency, front-end delivery, Python data pipelines and BI — and in the habit of reporting the result even when the result is negative.

EDUCATION

VIT-AP University, Amaravati *2024 – 2029*

Integrated M.Tech — Computer Science & Engineering (in collaboration with Virtusa) *CGPA: 7.55 / 10*

Coursework: *Advanced Programming, Data Structures & Algorithms, Database Management Systems, Software Engineering, Frontend Design & Testing*

SR Junior College — Intermediate (Class XII, MPC) *2022 – 2024 · 94.8%*

DAV Public School — Secondary School (Class X) *81.17%*

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Kotlin, SQL, Java, HTML, CSS
Backend & APIs: Node.js, Fastify, Express, FastAPI, REST design, JWT auth with rotating refresh tokens, RBAC, Zod validation, webhooks
Frontend: React 19, Next.js 15, Angular, Tailwind CSS, Jetpack Compose, progressive web apps, dependency-free canvas UI, accessibility (A11y)
Databases: PostgreSQL 16 + PostGIS, MySQL, Drizzle, Prisma, schema & index design, transactional concurrency control
Data & ML: pandas, NumPy, scikit-learn, Ultralytics YOLO11, ONNX, exploratory data analysis, walk-forward back-testing, conformal prediction, Monte Carlo simulation
Visualisation & BI: Power BI, Tableau, Matplotlib, Chart.js, Advanced Excel dashboards, data storytelling
Cloud, Tooling & QA: AWS, Docker & Compose, Git, GitHub Actions CI, Gradle, Vitest, ESLint, PyInstaller

SELECTED PROJECTS

Trade Research — Verifiable Research & Execution Platform *Aug – Sep 2026*

Stack: *Python, scikit-learn, pandas, FastAPI, PostgreSQL, React, MetaTrader 5, PyInstaller*

- Built a leakage-free walk-forward back-test benchmarking six scikit-learn models against naive and drift baselines, with conformal prediction intervals reporting honest held-out coverage, Monte Carlo price paths, and a cost-aware trading simulation reporting Sharpe and drawdown.
- Inverted the usual "AI analyst" design so Python computes and the model only interprets: indicators are calculated in code, financials come from SEC EDGAR XBRL with accession numbers, missing data stays null, and a verifier machine-checks every figure in the finished note back to the snapshot.
- Implemented the execution platform end to end — FastAPI, PostgreSQL and React wiring alerts through risk checks and an OMS to the broker — plus an autonomous MetaTrader 5 demo harness with scheduled wind-down.
- Reported the negative result rather than burying it: none of the strategies measured here has a demonstrable edge, and the README says so first.
- Measured the packaging decision instead of preferring one: PyInstaller onedir starts in 1,000 ms against onefile's 5,400 ms, so the 24-command toolkit ships as a directory.

Private repository — available on request

RoadAssist Bharat — Offline-First Emergency Mobility Platform *Aug – Sep 2026*

Stack: *TypeScript, Fastify, PostgreSQL 16 + PostGIS, Drizzle, Kotlin, Jetpack Compose, Python, YOLO11, Docker, GitHub Actions*

- Designed and built the entire platform solo across 117 commits: a 64-route Fastify modular monolith, a 56-table PostGIS schema with 138 indexes and 5 migrations, four browser surfaces, a native Android client and a computer-vision pipeline.
- Verified the API with 636 assertions across six suites, zero failures — including 74 security attacks that must each be refused (cross-tenant access, role escalation, SQL injection, forged and alg:none tokens, unsigned webhooks) and 75 concurrency tests.
- Settled every race in the database rather than by timing: SELECT ... FOR UPDATE on booking rows with offer expiry re-checked under the lock, a partial unique index making "paid once" true under simultaneous confirmations, and a hash-chained audit log that Postgres RULES make immutable.
- Trained four YOLO11 variants on RDD2022 across four countries: the best reaches a measured mAP50 of 0.471 (mAP50-95 0.226) on held-out validation at ~135 ms/image, and a separate nano tier trades accuracy for ~48 ms/image on CPU for Pi-class hardware. Potholes remain the hard class at ~0.24, and the per-class breakdown says so.

- Shipped Off-Grid Mode as architecture, not fallback: a Kotlin SOS ladder (data → SMS → 112 → offline queue) where exactly one channel owns an emergency, proven by 21 unit tests on that file alone, so a retry never dispatches a second ambulance.
- Wrote CI gates that fail the build when a document disagrees with the measured figures, when a file:line citation rots, or when module boundaries break — after the same wrong number reached twenty documents three times and a human caught it each time.
- Made a feature phone a first-class client: a complete booking over SMS with no app, in eight languages switched by text command — seven of them machine-translated and not yet native-reviewed, which the project states rather than implies.

github.com/saatwik-1157/roadassist-bharat

SkillForge — Entrepreneurship Enablement Platform *2026*

Stack: *TypeScript, Next.js 15, React 19, Express, Prisma, PostgreSQL 16, Tailwind CSS, Docker*

- Architected a full-stack monorepo turning skills into businesses: a Next.js 15 / React 19 front end over an Express + Prisma API on PostgreSQL 16, containerised with Docker Compose.
- Built a recommendation engine scoring 150+ business ideas against a user's skills, budget, available time and preferred business type, backed by a business-readiness assessment.
- Implemented production-grade auth — JWT with rotating refresh tokens, OTP email verification, password reset, Google login and role-based access control across entrepreneur, mentor and admin roles.
- Shipped the surrounding product: mentor booking with reviews, courses with auto-issued certificates, a threaded community forum, notifications, and an admin console for content approval and analytics.

github.com/saatwik-1157/skillforge

MachineSense AI — Explainable Industrial Intelligence *Smart India Hackathon 2026*

Stack: *Python, FastAPI, JavaScript, Canvas*

- Built an Industry 4.0 prototype for Indian industrial units: a FastAPI backend simulating a fleet of IoT-instrumented machines (vibration, temperature, current, sound, power) behind an analytics engine and a six-view real-time dashboard.
- Implemented the AI layer — a weighted 0-100 machine health score, MAD z-score anomaly detection, remaining-useful-life and failure-probability forecasting, and energy/cost/CO2 optimisation across the fleet.
- Made every score explainable: each health number is decomposed by sensor with plain-language root-cause reasoning, and a digital twin simulates a maintenance action before anyone touches the real machine.
- Mirrored the same scoring logic in dependency-free browser JavaScript so the demo behaves identically with the backend down, and labelled on screen which mode is running.

github.com/saatwik-1157/machinesense-ai

Founder Labs Autopilot — Ads → Voice → Revenue Closed Loop *Sep 2026*

Stack: *TypeScript, Node 24, Meta Ads API, Anthropic SDK, webhooks*

- Implemented an AI orchestrator that picks a niche, writes the offer and creative, publishes a guarded Meta test campaign, hands each lead to a voice agent, reads structured call outcomes back, and decides KEEP / KILL / ITERATE / SCALE from revenue rather than clicks.
- Carried campaign, adset, ad and creative identifiers from the lead form through the voice agent's call metadata and back on the post-call webhook — the round trip that lets a sale be attributed to the exact hook that paid for it.
- Put spend behind guardrails and two human approval gates, and made the whole loop runnable against mock providers so it demonstrates end to end with no ad spend and no phone calls.

github.com/saatwik-1157/ads-voice-revenue-loop

Smart University Management System (SUMS) *2026*

Stack: *MySQL, JavaScript, Chart.js, CSS3*

- Designed a normalised MySQL schema for an enterprise university system managing students, courses, faculty and academic records, with the query layer behind it.
- Built a glassmorphism front end with Chart.js dashboards surfacing enrolment and performance insights over the live data.

github.com/saatwik-1157/database_management_SUMS

OfferPilot — Job-Application Platform for F-1 / OPT Students *2026*

Stack: *JavaScript, HTML, CSS, localStorage*

- Built a dependency-free static platform — marketing site, onboarding, client dashboard, profile, legal pages and a passcode-gated ops console — with no build step and full localStorage persistence.
- Shipped the RezForge resume-tailoring engine, an interview kanban, a referral programme and analytics charts on a hand-written SVG engine, plus an OPT Runway Calculator computing work-authorisation runway, unemployment-day budget and H-1B lottery timing.
- Labelled the prototype honestly on the page: the match and resume engines are deterministic mocks and no real applications are submitted.

github.com/saatwik-1157/offerpilot

Angular Applications — ShopCart, Notes, Search & Auth *2026*

Stack: *Angular 22, TypeScript 6, standalone components, signals, Vitest*

- **ShopCart (e-commerce):** searchable, category-filtered catalogue, product detail pages, a quantity-aware cart with computed totals, and signup/sign-in with a route guard protecting checkout — 14 Vitest unit tests across cart, accounts and header.

- **Search template:** a user list loaded from JSON through a service and filtered on screen as the query is typed, matching on either name or email address.
- **Notes app:** add, update and delete notes held in a service-owned signal, rendered through separate form, list and single-note components.
- **Authentication app:** a login page validated in a service, with the home page reachable only through a route guard after a successful sign-in.
- Every experiment's lab record is generated from the live source on disk rather than written by hand, so the submitted document cannot disagree with the code that ran.

ExamPlanner — Android Study Planner *2026*

Stack: *Kotlin, Android, Java*

- Built a native Android study planner with authentication, exam and schedule management, and a Pomodoro timer with customisable focus and break intervals.

GenAI-Powered Data Analytics — Job Simulation *Forage · Jun 2026*

Stack: *Python, pandas, GenAI, data storytelling*

- Performed exploratory data analysis and risk profiling on a real-world lending dataset, built a model to predict customer delinquency, and delivered the data storytelling behind an AI-driven collections strategy.

Additional projects

- PathAhead — production-ready responsive site for a career-guidance platform running workshops for intermediate students across India (41 commits).
- Gemini Chat Client — a complete browser-only chat client: multi-thread sidebar, four personas, vision and voice input, Markdown rendering sanitised with DOMPurify, localStorage persistence, no server and no build step.

EXPERIENCE & LEADERSHIP

Campus Ambassador — SmartED *Jun 2026*

- Led campus outreach for SmartED's initiatives; formally recognised for leadership, dedication and the connections built across the student community.

Data Analytics Job Simulations — Deloitte, Quantium & GenAI programmes via Forage *Jun – Aug 2026*

- Deloitte (Jul 2026): completed practical tasks in data analysis and forensic technology.
- Quantium (Aug 2026): data preparation and customer analytics, experimentation and uplift testing, and the commercial application of the results.
- GenAI-Powered Data Analytics (Jun 2026): exploratory data analysis and risk profiling on a lending dataset, delinquency prediction, and the data storytelling behind an AI-driven collections strategy.

CERTIFICATIONS

- Google Analytics Certification — Google, May 2026 – May 2027 (ID 183723043)
- GenAI-Powered Data Analytics (Job Simulation) — Forage, Jun 2026
- Data Analytics Job Simulation — Deloitte via Forage, Jul 2026
- Data Analytics Job Simulation — Quantium via Forage, Aug 2026
- Introduction to AWS — AWS Training & Certification, May 2026
- AWS Certified Cloud Practitioner (CLF-C02) Domain 1 Review — AWS, May 2026
- AWS Certified Data Engineer – Associate (DEA-C01) Exam Prep — AWS, May 2026
- Exploring SAP Analytics Cloud — SAP Learning, Jun 2026
- Becoming an SAP Data Architect — SAP Learning, Jun 2026

ACHIEVEMENTS

- Sole author of RoadAssist Bharat — 117 commits covering API, Android client, CV pipeline, schema, six test suites and CI, with 636 assertions passing over the API and zero failures.
- Trained and measured four YOLO11 detector variants on RDD2022, reaching mAP50 0.471 on held-out validation and publishing the per-class breakdown including the classes that perform badly.
- Earned nine professional certifications across data analytics and cloud — AWS, Google and SAP — including three Forage job simulations (Deloitte, Quantium, GenAI analytics).
- Participant, HackDevengers 1.0 hackathon — organised by Devengers, 2026.
- Scored 94.8% in Class XII and 81.17% in Class X; recognised as Campus Ambassador, SmartED (2026).
- Designed, built and deployed a dependency-free personal portfolio at saatwik-1157.github.io.